

1. Artificial Intelligence (AI)

The simulation of human intelligence processes by machines, especially computer systems. This includes learning, reasoning, and self-correction.

2. Machine Learning (ML)

A subset of AI that enables systems to learn and improve from experience without being explicitly programmed. Models improve automatically through data exposure.

3. Deep Learning

A type of machine learning using multi-layered artificial neural networks to learn hierarchical representations of data. Powers many modern breakthroughs in vision and language.

4. Neural Network

A computational model inspired by the human brain, consisting of interconnected nodes (neurons) organized in layers that process and transmit information.

5. Supervised Learning

Training a model on labeled data, where the correct output is known. The model learns to map inputs to the correct outputs (e.g., image classification).

6. Unsupervised Learning

Training on unlabeled data to find hidden patterns or structures. Common uses include clustering and dimensionality reduction.

7. Reinforcement Learning

A learning paradigm where an agent learns to make decisions by interacting with an environment, receiving rewards or penalties for actions.

8. Natural Language Processing (NLP)

A field of AI focused on enabling computers to understand, interpret, and generate human language in a useful way.

9. Large Language Model (LLM)

A deep learning model trained on vast amounts of text data, capable of understanding and generating human-like text (e.g., Grok, GPT models).

10. Overfitting

When a model learns the training data too well, including noise and specifics, resulting in poor performance on new, unseen data.

11. Gradient Descent

An optimization algorithm used to minimize the loss function by iteratively adjusting model parameters in the direction of the steepest descent.

12. Tokenization

The process of breaking text into smaller units (tokens) such as words, subwords, or characters, which models can process.

13. Hallucination

When an AI model generates plausible-sounding but factually incorrect or fabricated information.

14. Transformer

A neural network architecture (introduced in 2017) that relies on attention mechanisms. It forms the foundation of most modern LLMs.

15. Embedding

A numerical vector representation of data (words, images, etc.) in a high-dimensional space that captures semantic meaning and relationships.